

MacSync-WT1 Wi-Fi Temperature & Humidity Sensor Datasheet

Product Overview

MacSync-WT1 is a compact, **Wi-Fi-based temperature and humidity sensor with an inbuilt battery**, designed for accurate indoor and semi-industrial environmental monitoring where direct IP connectivity is preferred. It enables real-time climate data transmission over standard Wi-Fi networks without the need for gateways or long-range network infrastructure.

Engineered with industrial-grade sensing elements and optimized power management, MacSync-WT1 delivers fast response, low drift, and long-term measurement stability while operating on battery power using intelligent sleep-wake cycles. This makes it ideal for deployments where wired power is unavailable but reliable Wi-Fi coverage exists.

The device securely publishes temperature and humidity data to **local servers, on-premise dashboards, or cloud platforms** using standard protocols such as **MQTT, HTTP, and HTTPS**. Built-in BLE support allows easy provisioning, diagnostics, and configuration through the Macnman Maya mobile application.

Housed in a rugged, ingress-protected enclosure, MacSync-WT1 is well suited for **smart buildings, HVAC systems, cold rooms, warehouses, laboratories, healthcare facilities, and industrial indoor environments**, delivering low-latency, reliable Wi-Fi-based environmental monitoring with minimal maintenance.





Key Features

High-Accuracy Temperature & Humidity Sensing

Precision temperature and humidity measurements using industrial-grade sensing elements for reliable indoor climate monitoring.

Battery-Operated Design

Inbuilt industrial-grade battery enables flexible deployment where wired power is unavailable.

Standard IoT Protocol Support

Supports MQTT, HTTP, and HTTPS for seamless integration with dashboards, BMS, and IoT platforms.

Secure Wireless Communication

WPA2 / WPA3 Wi-Fi security with TLS-encrypted data transmission.

Compact & Lightweight Enclosure

Small form factor suitable for wall or ceiling mounting in offices, warehouses, and equipment rooms.

Smart Data Reporting

Supports time-based, threshold-based, and event-driven reporting to optimize battery usage.

Low-Power Wi-Fi Operation

Optimized Wi-Fi communication with intelligent sleep-wake cycles to enable long battery life.

Direct IP Connectivity

Sends data directly to local servers or cloud platforms over Wi-Fi without requiring gateways.

Fast Response & Long-Term Stability

Low sensor drift and stable performance over long operating periods in indoor and semi-industrial environments.

Local Configuration via BLE

Easy provisioning, diagnostics, and configuration using the Macnman Maya mobile application.

Ingress-Protected Housing

IP-rated enclosure designed for reliable operation in industrial and commercial indoor environments.

Smart Building & HVAC Ready

Ideal for comfort monitoring, energy optimization, and indoor environmental intelligence.

Suited Applications

Smart Buildings & Offices

Continuous indoor temperature and humidity monitoring for occupant comfort, energy optimization, and building management systems.

Cold Rooms & Cold Storage

Reliable environmental monitoring for pharmaceutical storage, vaccine cold rooms, food storage, and cold-chain facilities.

Laboratories & Clean Rooms

Accurate climate monitoring for research labs, testing facilities, calibration rooms, and controlled environments.

Data Centers & Server Rooms

Monitors temperature and humidity to protect critical IT infrastructure and ensure operational reliability.

Industrial Indoor Areas

Monitors environmental conditions inside factories, production floors, control rooms, and utility spaces with Wi-Fi coverage.

HVAC Monitoring & Optimization

Real-time temperature and humidity tracking across HVAC zones to improve efficiency, performance, and preventive maintenance.

Warehouses & Storage Facilities

Ensures controlled temperature and humidity conditions for FMCG, electronics, raw materials, and finished goods.

Hospitals & Healthcare Facilities

Supports temperature and humidity monitoring in wards, pharmacies, equipment rooms, and medical storage areas.

Retail & Commercial Spaces

Maintains indoor comfort and climate conditions in malls, stores, hotels, and commercial buildings.

Technical Specifications

General Specifications

Part Order Number	MacSync-WT1
Product Brand	Macnman Technologies Pvt Ltd

Temperature Measurement

Range	-30°C to +70°C
Accuracy	0°C to +70°C: ±0.3°C & -30°C to 0°C: ±0.6°C
Resolution	0.1°C
Calibration	Factory calibrated
Probe Options	Internal & External Both Options

Humidity Measurement

Range	0% to 100% RH
Accuracy	10% to 90% RH: $\pm 3\%$ RH & <10% or >90% RH: $\pm 5\%$ RH
Resolution	0.5% RH
Calibration	Factory calibrated

Pressure Measurement

Range	300 hPa to 1100 hPa
Accuracy	± 1.0 hPa (typical at 25°C, 300–1100 hPa)
Resolution	0.1 hPa
Long-Term Stability	$\leq \pm 1$ hPa/year
Calibration	Factory calibrated

Wireless Communication

Wi-Fi Standards	IEEE 802.11 b/g/n/ac
Frequency Bands	2.4 GHz & 5 GHz (Dual-Band)
Operating Modes	Station Mode (Client) Access Point Mode (Configuration)
Network Protocols	MQTT, HTTP, HTTPS
Security	WPA2-PSK / WPA3 TLS / SSL Encrypted Communication
IP Configuration	DHCP / Static IP
Typical Indoor Range	Up to 100 m (environment dependent)

LED Status

LED Type	1 Internal Multi-Color LED
Solid Blue	Device is connected with Maya App
Flashing Blue	OTA (Over-the-Air) update in progress
Slow Blinking Green	Attempting to connect to the network

Single Red Blink	A confirmed parameter of MacSync is being modified
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Device Configuration

Configuration	Via Macnman Maya Android App
Uplink Modes	Time Based / Threshold Based / Sampling Based
Stored Records*	30,000 Records
Remote Configuration	JSON Format Data - from server

Supported Cloud & Platforms

Deployment Models	On-Premise Servers Private Cloud Deployments Public Cloud Platforms
IoT Cloud Platforms	AWS IoT Core Microsoft Azure IoT Hub Google Cloud (via MQTT / HTTPS) Custom IoT Platforms
Integration Methods	MQTT Brokers REST / HTTPS APIs Webhooks
Enterprise Systems	SCADA Systems BMS / IBMS Platforms Data Lakes & Analytics Engines

Physical Characteristics

Power Supply	9-36 V DC
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)
Ingress Protection	IP65 standard (IP67 optional variant)
Housing Material	Glass filled Nylon
Color	Matte White
Weight	~110g
Mounting Options	Pole Mounting / Wall Mounting

Electrical Characteristics

Input Voltage Range	9-36 V DC (External Supply)
Power Consumption	~100 µA in sleep mode ~150 mA during transmission
Input Impedance	>10 MΩ for voltage inputs

Communication Options

BLE Connectivity	For local configuration and diagnostics via Maya App
Cloud Protocols	MQTT / HTTPS
Serial Interface	RS485 Modbus RTU support (optional)

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

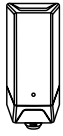
Data Encryption	AES-256 end-to-end encryption
Authentication	OTAA activation modes
Firmware Protection	Secure boot and signed OTA updates

MacSync-WT1 Variants

VARIANT NAME	POWER SOURCE	BATTERY CAPACITY	SOLAR SUPPORT	EXTERNAL FLASH	IDEAL FOR
MacSync-WT1-PO	12-24 V (Power)				Industrial or commercial sites with stable power supply
MacSync-WT1-BO	Battery	19 Ah			Remote or portable deployments with low power

What's in the Box?

Make sure box contains following items before installations



MacSync



**Mounting
Clip**



**User manual &
Warranty Card**



**Junction
Box***



**Mounting
Screws**

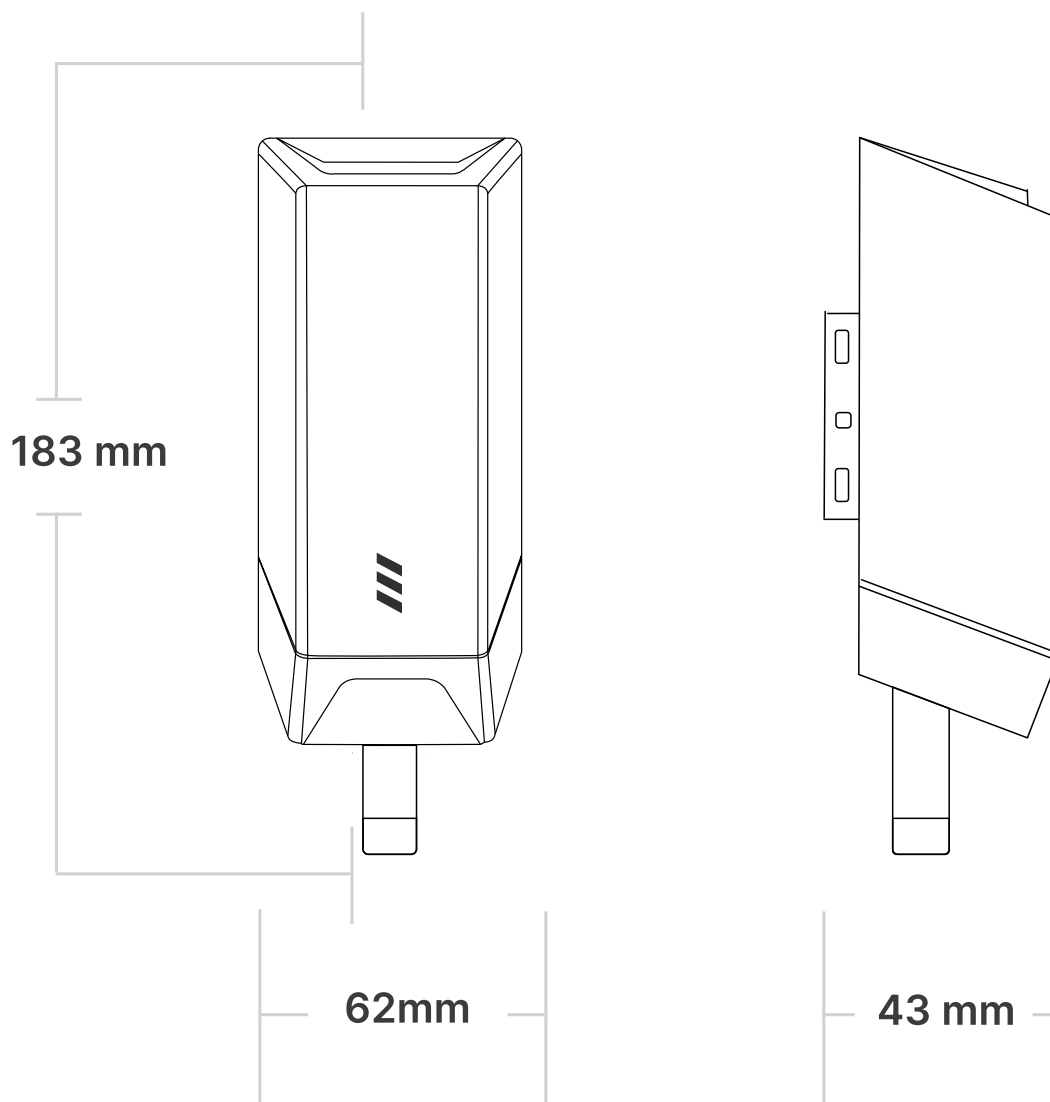


**Power
Adapters***

** Note**

Accessories must be purchased separately & may not be included in the box.

MacSync-WT1 Dimensions



Frequently Asked Questions

What is the MacSync-WT1 used for?

MacSync-WT1 is a compact Wi-Fi temperature and humidity sensor with an inbuilt battery for accurate indoor and semi-industrial environmental monitoring. Typical applications include Smart Buildings & Offices, HVAC Monitoring & Optimization, Cold Rooms & Cold Storage, Warehouses & Storage Facilities, Laboratories & Clean Rooms and Hospitals & Healthcare Facilities.

What does the MacSync-WT1 measure, and how accurately?

Measurement specifications of the MacSync-WT1:

- Temperature: range -30°C to +70°C, accuracy 0°C to +70°C: $\pm 0.3^{\circ}\text{C}$ & -30°C to 0°C: $\pm 0.6^{\circ}\text{C}$, resolution 0.1°C.
- Humidity: range 0% to 100% RH, accuracy 10% to 90% RH: $\pm 3\%$ RH & <10% or >90% RH: $\pm 5\%$ RH, resolution 0.5% RH.
- Pressure: range 300 hPa to 1100 hPa, accuracy ± 1.0 hPa (typical at 25°C, 300–1100 hPa), resolution 0.1 hPa.

How does the MacSync-WT1 connect and send data over Wi-Fi?

The MacSync-WT1 connects over IEEE 802.11 b/g/n/ac Wi-Fi on 2.4 GHz & 5 GHz (Dual-Band). Security: WPA2-PSK / WPA3, TLS / SSL Encrypted Communication. IP configuration: DHCP / Static IP. It sends data using MQTT, HTTP, HTTPS. Typical indoor range is up to 100 m (environment dependent).

What is the IP rating and operating temperature of the MacSync-WT1?

The MacSync-WT1 has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

How is the MacSync-WT1 configured?

The MacSync-WT1 is configured with the Macman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

Which cloud platforms does the MacSync-WT1 work with?

The MacSync-WT1 can send data to platforms including AWS IoT Core, Microsoft Azure IoT Hub, Google Cloud (via MQTT / HTTPS), Custom IoT Platforms. Deployment models: On-Premise Servers, Private Cloud Deployments, Public Cloud Platforms. Integration methods: MQTT Brokers, REST / HTTPS APIs, Webhooks.

Related Products and Resources

- Related datasheets: [MacSync LTS X1 LoRaWAN temperature and humidity sensor](#), [MacSync-WRTDX2 Wi-Fi RTD PT100 temperature sensor](#), [MacSync-WRTDX5 Wi-Fi RTD PT1000 temperature sensor](#), [MWTH 301 compact Wi-Fi temperature and humidity sensor](#)
- Product page: [MacSync-WT1 on macman.com](#)
- Help: [Warranty terms and conditions](#) and [contact Macman support](#)